

# Product Manager Assignment

## E-commerce System MVP Design & Basic Technical Understanding

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Position: Product Manager

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Submission Package: GitHub Repository + PDF/DOCX

### Executive Summary

This MVP design focuses on launching a reliable end-to-end transaction flow before adding advanced growth features. The proposed scope prioritizes product browsing, cart/order submission, payment, shipment, inventory control, and order lifecycle management. The solution also defines key exception handling, order status transitions, backend validation principles, and launch metrics so the product can be operated safely after release.

| Area                | Design Choice                                                                                                                                 |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| MVP Goal            | Enable customers to complete the core flow: browse, add to cart or buy now, submit order, pay, receive shipment, and complete order.          |
| Product Principle   | Prioritize transaction reliability, inventory accuracy, clear status transitions, and basic operations efficiency.                            |
| Out of MVP          | Coupons, recommendation engine, loyalty points, advanced analytics, multi-merchant settlement, and complex refund automation.                 |
| Technical Principle | Frontend display improves UX, but backend validation is the final source of truth for stock, price, product status, payment, and order state. |

### Submission Structure

- 1. User Roles and Core Scenarios
- 2. MVP Feature List and Priority
- 3. Core Process Design
- 4. Simplified PRD - Product Detail Page
- 5. API Understanding Answers
- 6. Code Logic Reading Answers
- 7. Order Status Understanding Answers
- 8. Bug / Exception Analysis Answers
- 9. Data Metrics Design

## 1. User Roles and Core Scenarios

| Role                              | Primary Goal                                                      | Core Scenarios                                                                                                                                                                                   |
|-----------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Customer / Buyer                  | Find products and complete purchases with clear order visibility. | Register/login, browse products, search/filter, view product details, select SKU, add to cart, buy now, submit order, pay, view order status, cancel unpaid order, request refund when eligible. |
| Operations Staff / Merchant Staff | Maintain sellable products and process orders accurately.         | Create/edit products, manage SKU price and inventory, publish/unpublish products, review paid orders, arrange shipment, update tracking number, handle abnormal orders and refund requests.      |
| Platform Admin                    | Control platform permissions and monitor operational risk.        | Manage staff permissions, view platform-level data, audit operation logs, handle exceptional orders, configure basic rules such as payment timeout and refund policy.                            |

## 2. MVP Feature List and Priority

**Priority definition:** P0 = must-have for core transaction flow; P1 = should-have for user experience or operating efficiency; P2 = later iteration.

| Module             | Feature                          | Priority | Reason                                                                                                                    |
|--------------------|----------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------|
| Customer Side      | Login / Registration             | P1       | Useful for order history and identity, but guest browsing can still work. Login becomes required before order submission. |
| Customer Side      | Product List                     | P0       | Customers need an entry point to browse available products.                                                               |
| Customer Side      | Product Detail Page              | P0       | Customers need product information, price, SKU, stock, images, and purchase actions before buying.                        |
| Customer Side      | Product Search / Category Filter | P1       | Improves discovery and efficiency, but not required for a small MVP catalog.                                              |
| Customer Side      | Shopping Cart                    | P0       | Supports multi-item purchase and standard checkout.                                                                       |
| Customer Side      | Buy Now                          | P1       | Improves conversion for single-item purchase, but cart-based checkout can cover the basic flow.                           |
| Customer Side      | Submit Order Page                | P0       | Collects address, SKU quantity, price confirmation, and creates an order.                                                 |
| Customer Side      | Payment                          | P0       | Required to complete the transaction and move order to paid status.                                                       |
| Customer Side      | Order List / Order Detail        | P0       | Customers need to track payment, shipment, and completion status.                                                         |
| Customer Side      | Cancel Order                     | P1       | Required for unpaid order management and reducing support pressure.                                                       |
| Customer Side      | Refund Request                   | P1       | Important after payment, but can start with simple manual review in MVP.                                                  |
| Admin / Operations | Product Management               | P0       | Products must be created, edited, and published before sale.                                                              |
| Admin / Operations | SKU / Inventory Management       | P0       | Inventory must be maintained at SKU level to prevent overselling.                                                         |
| Admin / Operations | Order Management                 | P0       | Staff need to review orders and process abnormal states.                                                                  |
| Admin / Operations | Shipment Management              | P0       | Paid orders need shipment information and tracking updates.                                                               |
| Admin / Operations | User Management                  | P1       | Needed for internal permission control, but can be simple in MVP.                                                         |
| Admin / Operations | Data Dashboard                   | P2       | Useful for management, but not required for the first transaction loop.                                                   |
| Platform           | Operation Logs                   | P2       | Useful for audits and troubleshooting, can be expanded after launch.                                                      |

## 3. Core Process Design

### 3.1 Customer Order and Payment Process

| Step                    | Normal Flow                                                                                                  | Status / Data Change                                                                                 |
|-------------------------|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| 1. Select product/SKU   | Customer opens product detail page, selects SKU and quantity, then chooses Add to Cart or Buy Now.           | No order is created yet. Frontend displays selected SKU price and stock.                             |
| 2. Submit order         | System checks login, address, product status, SKU stock, price, and quantity. If valid, it creates an order. | Order status = PENDING_PAYMENT. SKU stock can be locked or deducted according to inventory strategy. |
| 3. Payment              | Customer pays through supported payment channel. Payment callback confirms success.                          | Order status = PAID. Payment transaction ID and paid time are recorded.                              |
| 4. Merchant fulfillment | Operations staff reviews paid order and arranges shipment.                                                   | Order remains PAID until shipment is confirmed.                                                      |
| 5. Shipment             | Staff enters logistics provider and tracking number.                                                         | Order status = SHIPPED. Shipment time is recorded.                                                   |
| 6. Completion           | Customer confirms receipt or system auto-completes after a defined period.                                   | Order status = COMPLETED. Completion time is recorded.                                               |

| Exception Scenario                 | Handling Rule                                                                                                                                      | Expected Result                                                                                     |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Insufficient inventory             | Backend re-checks SKU stock during order submission. If stock is insufficient, order creation is blocked.                                          | User sees: "Insufficient inventory. Please reduce quantity or choose another SKU."                  |
| Product off sale but still in cart | Cart can display the item as invalid, but order submission must block it.                                                                          | User sees: "This product is no longer available." The invalid item cannot be selected for checkout. |
| Payment failure                    | Keep order as pending payment and allow the user to retry within payment time limit.                                                               | Order remains PENDING_PAYMENT. Payment failure reason is shown when available.                      |
| Payment timeout                    | If user does not pay before timeout, system cancels the order and releases locked stock.                                                           | Order status = CANCELLED. Reason = payment timeout.                                                 |
| Price changed before payment       | Backend validates latest price before order creation. If price changes after cart entry, the order page must show updated price before submission. | User must confirm the latest price before order creation.                                           |

**Recommended inventory strategy:** For MVP, lock stock after order submission and release it after payment timeout or cancellation. Final deduction can occur after payment success. This reduces overselling while protecting unpaid inventory from being locked forever.

### 3.2 Product Creation and Publishing Process

| Step                    | Normal Flow                                                                                                  | Status / Data Change                                                  |
|-------------------------|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| 1. Create draft         | Operations staff enters product name, category, description, images, SKU values, prices, and inventory.      | Product status = DRAFT. Product is not visible to customers.          |
| 2. Validate information | System checks mandatory fields, price format, SKU completeness, image existence, and non-negative inventory. | If validation fails, product remains DRAFT with error messages.       |
| 3. Save product         | Staff saves product information.                                                                             | Product details and SKU records are stored.                           |
| 4. Publish product      | Staff clicks Publish after required data is complete.                                                        | Product status changes to ON_SALE and becomes visible to customers.   |
| 5. Edit after publish   | Staff can edit price, images, description, and stock under permission rules.                                 | Changes are recorded. Stock and price changes affect new orders only. |

| Exception Scenario         | Handling Rule                                                                                            | Expected Result                                                                         |
|----------------------------|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Missing required field     | Block publish and highlight missing fields such as product name, image, price, SKU, or stock.            | Product remains DRAFT.                                                                  |
| SKU price or stock invalid | SKU price must be greater than 0; stock cannot be negative.                                              | Save/publish is blocked until corrected.                                                |
| Inventory set to 0         | Allow publish if business permits, but customer page must show out of stock and disable purchase button. | Product can be visible but cannot be purchased.                                         |
| Product taken off sale     | Staff changes status from ON_SALE to OFF_SALE.                                                           | Product is hidden from normal listing; existing cart items become invalid for checkout. |
| Published product in cart  | If product becomes OFF_SALE after being added to cart, backend blocks checkout.                          | User must remove the item or choose another available product.                          |

## 4. Simplified PRD - Product Detail Page

| Item                       | Content                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature Background         | The Product Detail Page is the main decision page before purchase. It must present product information clearly and connect the customer to Add to Cart or Buy Now while preventing invalid purchases.                                                                                                                                                                              |
| User Goal                  | Customers need to understand the product, compare SKU options, see accurate price and stock, and start checkout confidently.                                                                                                                                                                                                                                                       |
| Page Fields                | Product name, product images, selling price, original price, discount display if applicable, sales count, product status, SKU options, selected SKU price, selected SKU stock, quantity selector, product description, delivery/support notes, Add to Cart button, Buy Now button.                                                                                                 |
| Business Rules             | Only ON_SALE products can be purchased. Purchase buttons are disabled when product status is OFF_SALE, selected SKU stock is 0, or quantity exceeds stock. Displayed price and stock must update after SKU selection. The backend must validate product status, SKU stock, price, and quantity again during order submission.                                                      |
| Error / Exception Handling | OFF_SALE: "This product is currently unavailable." Out of stock: "Out of stock." Quantity exceeds stock: "Only X items available." SKU not selected: "Please select a product option." Price updated: "Product price has changed. Please confirm before ordering."                                                                                                                 |
| Acceptance Criteria        | 1) Customer can view product name, images, price, SKU options, stock, and sales count. 2) SKU selection updates displayed price and stock. 3) Add to Cart and Buy Now work only when product is ON_SALE and selected SKU has enough stock. 4) OFF_SALE or out-of-stock products cannot be purchased. 5) Backend validation prevents invalid orders even if frontend data is stale. |

## 5. API Understanding Answers

| Question                        | Answer                                                                                                                                                                                                                                                                                                        |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. What can be displayed?       | Product ID, product name, selling status, selling price, original price, total stock, sales count, product image(s), SKU options, SKU-specific price, and SKU-specific inventory.                                                                                                                             |
| 2. How to display price: 19900? | It should be displayed as 199.00 yuan/dollars depending on the currency setting, because prices are commonly returned in cents/minor units. The frontend should format it as ¥199.00 if the platform currency is RMB.                                                                                         |
| 3. If status = OFF_SALE?        | The page may display product information for reference, but Add to Cart and Buy Now should be disabled. The page should show a clear message such as "This product is currently unavailable."                                                                                                                 |
| 4. If total stock is 0?         | The purchase button should be disabled and shown as "Out of Stock". Quantity selector should not allow purchase quantity above available stock.                                                                                                                                                               |
| 5. If user selects Black SKU?   | The displayed price should change to 20900 minor units, displayed as ¥209.00. The displayed inventory should change to 15.                                                                                                                                                                                    |
| 6. Missing fields               | Currency: needed to format price correctly. Product description/specifications: needed for customer decision-making. Category/brand: needed for navigation and search. Shipping information: needed for delivery expectation. Purchase limit/after-sales policy: needed to enforce rules and reduce disputes. |

## 6. Code Logic Reading Answers

| Question                                               | Answer                                                                                                                                                                                                                                                               |
|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. What is this function used for?                     | It determines whether a customer can submit an order for a selected product/SKU and purchase quantity.                                                                                                                                                               |
| 2. When can the user not submit?                       | When the product is not ON_SALE, when SKU stock is 0 or below, or when the requested quantity is greater than available SKU stock.                                                                                                                                   |
| 3. productStatus = ON_SALE, skuStock = 3, buyCount = 5 | The function returns canSubmit = false, reason = "Purchase quantity exceeds available inventory".                                                                                                                                                                    |
| 4. Additional checks to add                            | Login status, valid selected SKU, valid delivery address, minimum/maximum purchase quantity, purchase limit per user, latest price verification, product availability by region, payment method availability, abnormal account/risk control, and cart item validity. |

## 7. Order Status Understanding Answers

| Question                                  | Answer                                                                                                                                                                                                                                                      |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Initial status after submit order      | PENDING_PAYMENT                                                                                                                                                                                                                                             |
| 2. Status after successful payment        | PAID                                                                                                                                                                                                                                                        |
| 3. Status after merchant ships order      | SHIPPED                                                                                                                                                                                                                                                     |
| 4. Status if not paid within time limit   | CANCELLED, with cancellation reason = payment timeout.                                                                                                                                                                                                      |
| 5. Statuses where refund can be requested | Usually PAID and SHIPPED. Some platforms also allow refund/return request after COMPLETED depending on after-sales policy. Refund request should not be allowed for PENDING_PAYMENT, CANCELLED, or already REFUNDED orders.                                 |
| 6. Status transition flow                 | Normal flow: PENDING_PAYMENT -> PAID -> SHIPPED -> COMPLETED. Unpaid exception: PENDING_PAYMENT -> CANCELLED. Refund flow: PAID/SHIPPED/COMPLETED -> REFUNDING -> REFUNDED. Refund rejected returns to the previous valid order status according to policy. |

## 8. Bug / Exception Analysis Answers

**Issue:** Customer sees inventory = 1 on the product detail page, but order submission returns "Insufficient inventory."

| Possible Cause                           | Explanation                                                                                      | Investigation Method                                                                                            |
|------------------------------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Cached product detail inventory          | The product detail page may show cached stock while backend stock has already changed.           | Check cache TTL, cache update mechanism, and compare cache value with inventory database value.                 |
| Another customer purchased first         | Stock was 1 when the first user viewed the page, but another user submitted or paid before them. | Check inventory operation logs, order creation time, stock lock records, and payment success time.              |
| SKU stock and product stock inconsistent | Product total stock may show 1, but selected SKU stock may be 0.                                 | Check SKU stock table, product aggregate stock calculation, and selected skuld in the order submission request. |
| Cart item became invalid                 | Product or SKU may have become OFF_SALE after being added to the cart.                           | Check product status, SKU status, and cart validation result during checkout.                                   |
| Order submission API re-checks stock     | Backend correctly blocks order because actual stock is insufficient at submission time.          | Review order submission logs and validation response. Confirm backend stock is final source of truth.           |
| Frontend/backend out of sync             | Frontend may not refresh latest product/SKU data before checkout.                                | Check whether order confirmation page calls latest price/stock API before submission.                           |

**Recommended handling:** Do not rely on frontend inventory display. The backend must validate product status, selected SKU stock, quantity, and latest price at order submission. If validation fails, return a clear message and refresh the cart/order confirmation page with the latest stock. For high-concurrency products, add stock locking, idempotent order creation, and inventory operation logs.

## 9. Data Metrics Design

| Metric                      | Meaning                                                     | Why It Matters                                                                                        |
|-----------------------------|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Product Detail Page Views   | Number of views on product detail pages.                    | Measures product exposure and traffic quality.                                                        |
| Add-to-Cart Rate            | Users who add to cart / users who view product details.     | Measures product attractiveness and purchase interest.                                                |
| Buy Now Click Rate          | Users who click Buy Now / product detail page viewers.      | Shows direct purchase intent and product-page conversion strength.                                    |
| Order Submission Rate       | Users who submit orders / users who reach cart or checkout. | Measures checkout flow effectiveness and identifies friction before payment.                          |
| Payment Success Rate        | Paid orders / submitted orders.                             | Measures whether payment flow works properly and whether payment failures are blocking revenue.       |
| GMV                         | Total paid order value.                                     | Measures sales scale and business performance.                                                        |
| Inventory Out-of-Stock Rate | Out-of-stock product or SKU views / total product views.    | Shows missed sales opportunities and inventory planning issues.                                       |
| Order Fulfillment Time      | Average time from payment success to shipment.              | Measures operations efficiency and customer delivery experience.                                      |
| Refund Rate                 | Refunded orders / paid orders.                              | Indicates potential product quality, description accuracy, logistics, or customer expectation issues. |